Graph-Based Classification of EEG Signals

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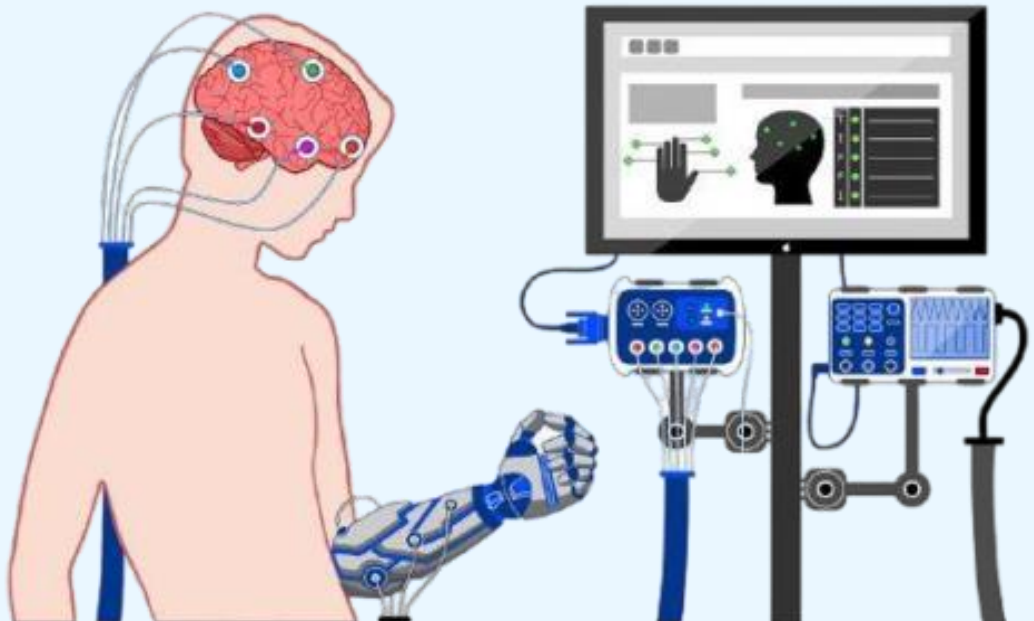
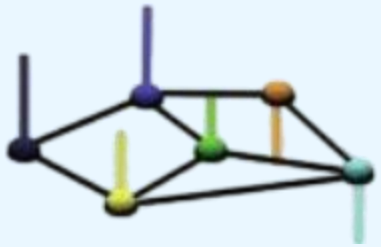
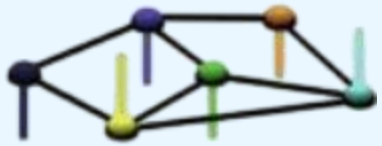
Introduction: Brain Computer Interfaces

- Command and control using brain signals (mostly EEG)
- Spellers, navigation, robot control, psychological disorders, emotion recognition, etc...
- Our purpose is to develop robust algorithms using graphs

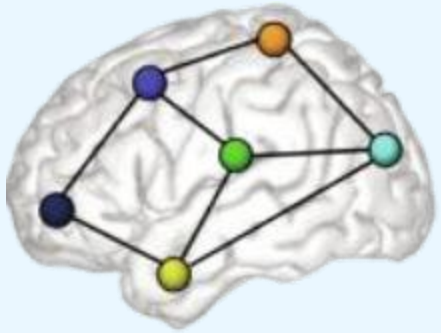


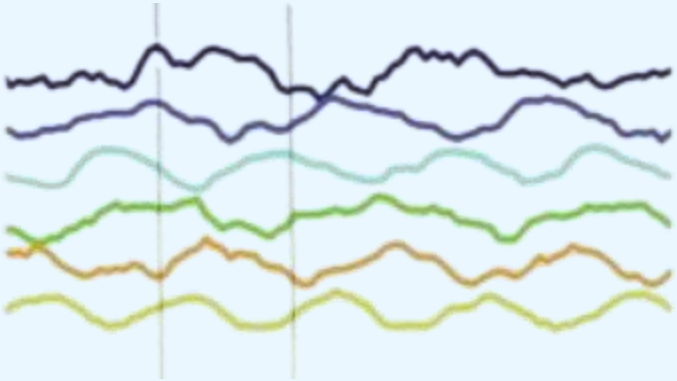
Brain Connectivity





- Structural Connectivity
  - Anatomical connection of neurons
- Functional Connectivity
  - Electrical activity between brain regions
- Effective Connectivity

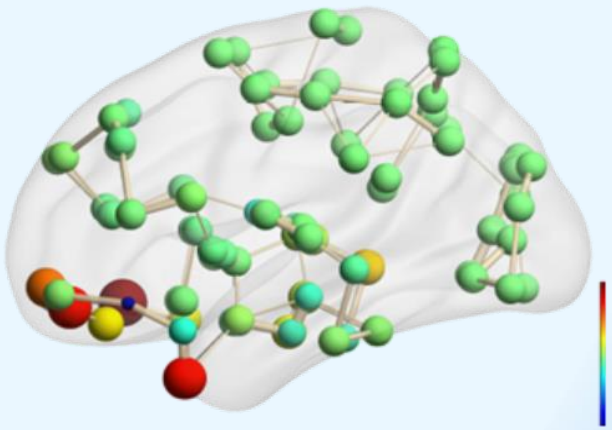




Graph Signal Processing

- Graph Fourier Transform and Graph Signal Filtering

$$\boldsymbol{\mathcal{L}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T \qquad \hat{\mathbf{f}} = \mathbf{U}^T\mathbf{f}$$



- Graph Smoothness and Total Variation

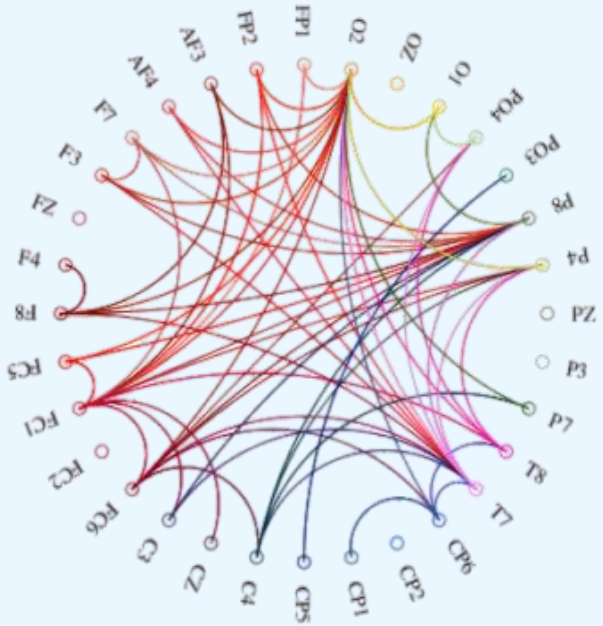
$$\mathrm{TV}\left(\mathbf{x}\right) := \mathbf{x}^\top \mathbf{L}\mathbf{x} = \sum_{i \neq j} W_{ij}\left(x_i - x_j\right)^2 = \sum_{k=1}^{N} \lambda_k \tilde{x}_k^2.$$

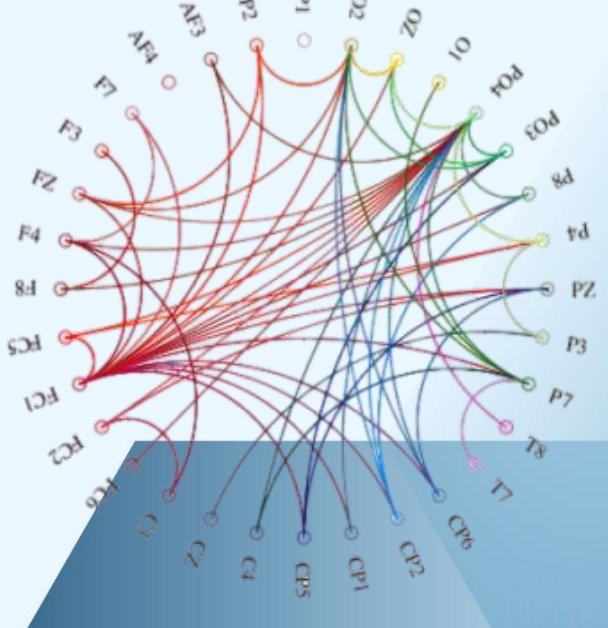
$$\min_W \sum_i \sum_j W_{ij} Z_{ij} - \alpha \sum_i \log(\sum_j W_{ij}) + \frac{\beta}{2}\|W\|_F^2$$

Discriminative Graph Learning

- Each graph is smooth to its signal and non-smooth to other signals

$$\min_{\mathbf{W}_c \in \mathcal{W}_m} \|\mathbf{W}_c \circ \mathbf{Z}_c\|_1 - \alpha \mathbf{1}^\top \log(\mathbf{W}_c \mathbf{1}) + \beta \|\mathbf{W}_c\|_F^2 - \gamma \sum_{k \neq c}^{C} \|\mathbf{W}_c \circ \mathbf{Z}_k\|_1,$$

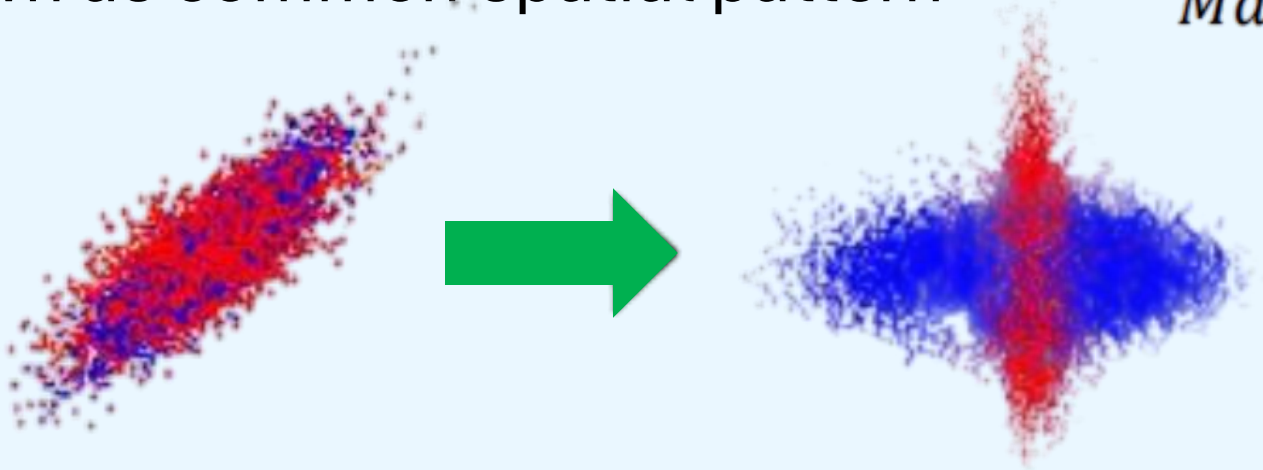




Fukunaga-Koontz Transform (FKT)

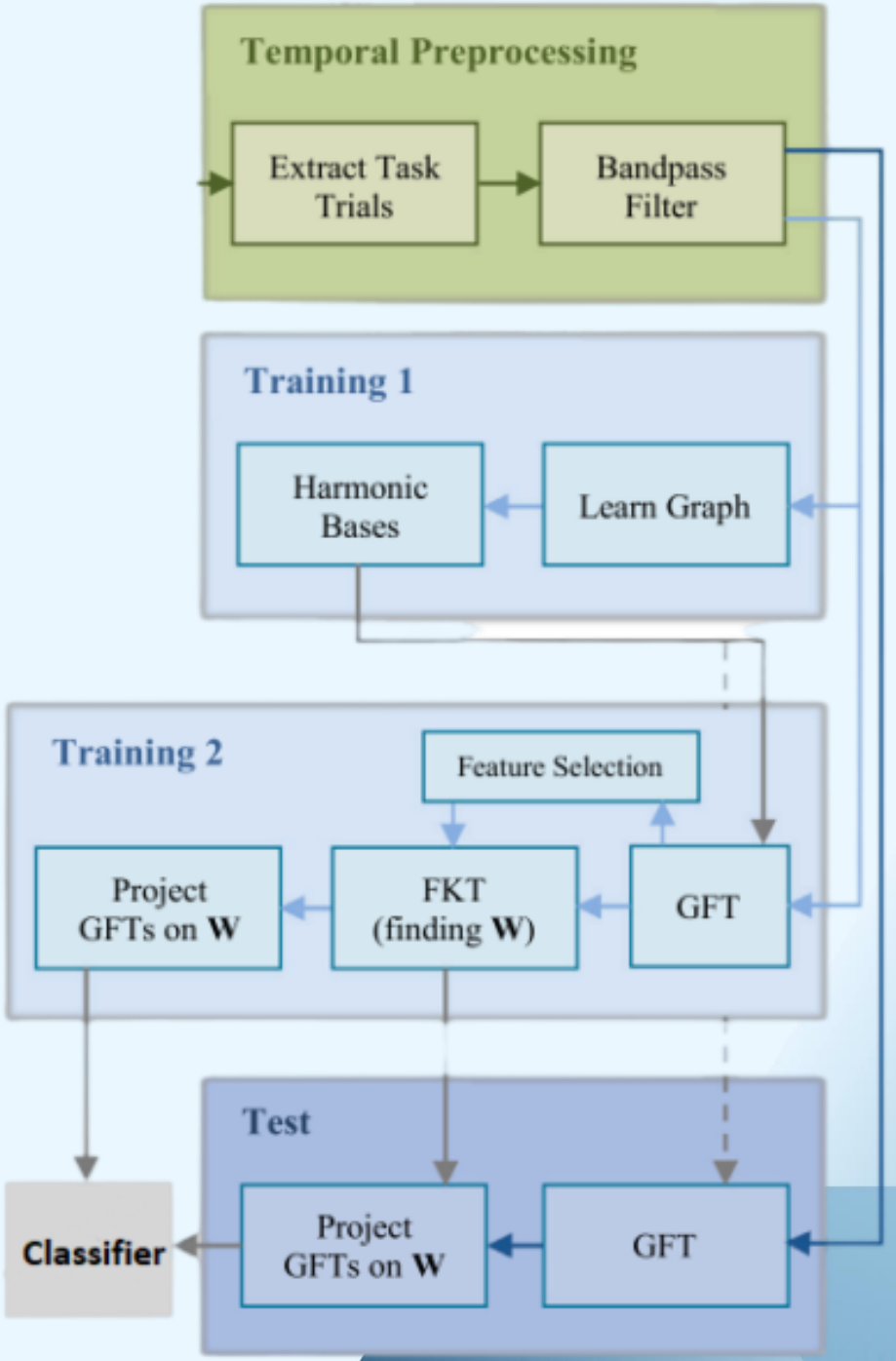
- Maximizing Rayleigh ratio
- Fukunaga-Koontz Transform (FKT) allows to find a discriminative subspace to project the GFT coefficients of the signals
- After transformation, first and last ***m*** variances are used for classification
- Also known as common spatial pattern

$$Max\ J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{C}_1 \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{C}_2 \boldsymbol{w}}$$



Framework

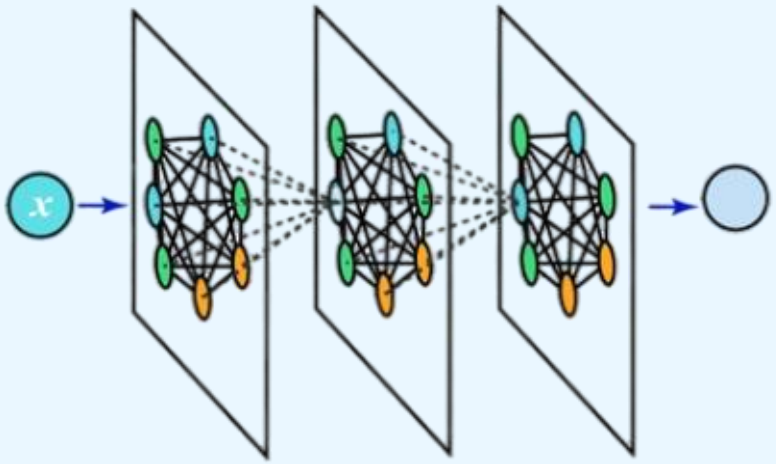
- Learning graphs of each class
- Applying Graph Fourier Transform on signals
- Using FKT algorithm and transforming the GFT coefficients
- Training a classifier on the first and last ***m*** variances



Graph Convolutional Neural Networks

- Graph-Temporal domain filtering using CNNS
- GFD: Graph Fisher Discriminant = Δ J
- Tensor-Train in fully connected layers to reduce parameters

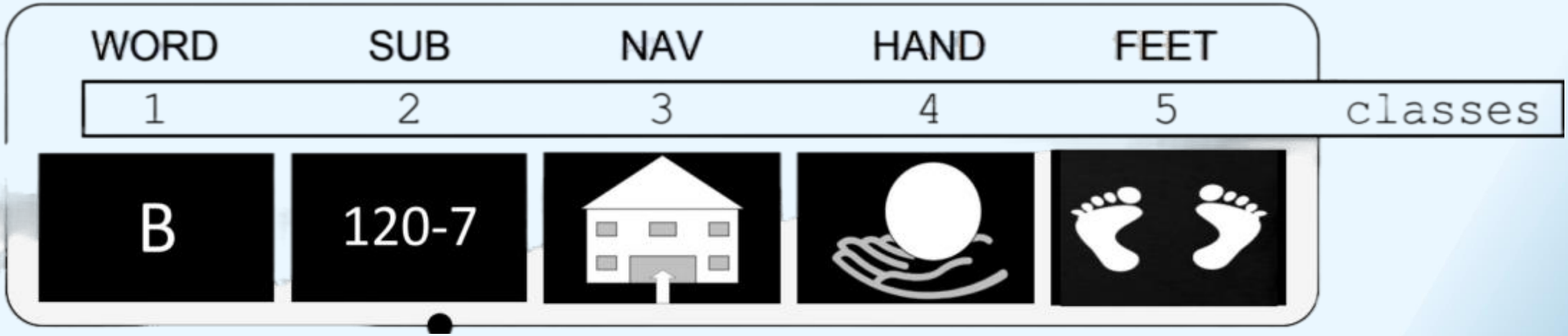
$$Loss = L_{Cross\ Entropy} + \lambda\, L_{GFD}$$



$$J(\boldsymbol{W}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}$$

Dataset

- 5 class data
- 30 channels and 200 data points
- Spinal cord injury and stroke



Results

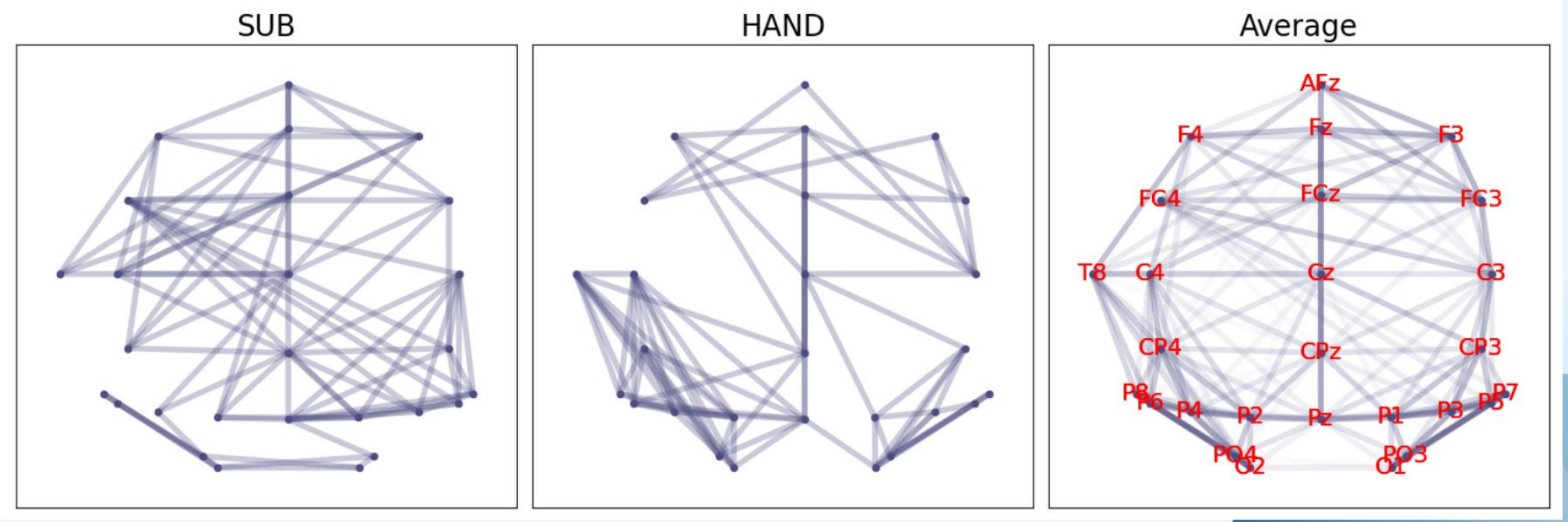
- Our model out-performed the models used in previous studies on 2 class data
- Accuracy on 5 class data can be improved by hyperparameter tuning or feature selection

Model	Accuracy (2 Class)
GFT + FKT (Learned Graphs)	88.5%
GFT + FKT (Structural Graphs)	85.0%
CSP + Trial Removal	82%

Model	Accuracy (5 class)
GTCNN + Learned Graphs	47%
GFT + FKT (Learned Graphs)	49%

Results

- Our results are consistent with previous studies
- During right hand movement left parietal regions are activated for planned movements while in subtraction frontal and higher order regions are involved
- Angular Gyrus includes frontal, temporal, and parietal areas in the left hemisphere. For language and arithmetic cognition



Summary

- We developed new algorithms based on graph learning and graph signal processing
- Results demonstrate high performance of graph-based methods

- Future Work
  - Improving model for 5 class data
  - Larger dataset for deep networks

References

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Any Questions?